DevOps Mastery

10-Day Intensive Training Program

Technology: DevOps | Duration: 10 day(s) | Mode: Online | Trainer: Vikram Chauhan

Program Overview

This 10-day DevOps program is generated by the Training TOC Agent using a structured domain curriculum. It combines concepts, daily labs, Agile/Jira practice, milestone reviews, and a final capstone for intermediate learners in Online mode.

Program Roadmap

Day 1: Linux Basics | Tools: Ubuntu + Bash | Jira: Update sprint board

Day 2: Networking Basics | Tools: curl + netstat + nmap | Jira: Log time on stories

Day 3: Docker Basics | Tools: Docker + Docker Hub | Jira: Move cards to In Progress/Done

Day 4: Jenkins CI/CD | Tools: Jenkins + Maven + Nexus | Jira: Add comments with build links

Day 5: Terraform IaC | Tools: Terraform + AWS + S3 | Jira: Sprint planning; review progress for Terraform IaC

Day 6: Kubernetes Fundamentals | Tools: kubectl + minikube + k9s | Jira: Log time on stories

Day 7: AWS Core | Tools: AWS CLI + AWS Console + Terraform | Jira: Move cards to In Progress/Done

Day 8: ELK Stack | Tools: ELK + Filebeat | Jira: Add comments with build links

Day 9: DevSecOps | Tools: Trivy + Snyk + AWS Secrets Manager | Jira: Update sprint board

Day 10: Capstone Project + Certification Roadmap | Tools: All DevOps Tools | Jira: Final sprint review, retrospective, release notes, and stakeholder demo

Prerequisites

- Laptop with required software access

- Basic computer and internet usage

- Interest in learning DevOps through practical labs

Learning Outcomes

- Understand DevOps concepts from foundation to implementation

- Complete daily hands-on labs and milestone assignments

- Use industry tools in realistic project workflows

- Track delivery using Agile/Jira practices

- Complete final capstone and certification roadmap review

Day 1: Linux Basics

Tools: Ubuntu + Bash | Jira Focus: Update sprint board

Morning Session: Linux Basics - Concepts (9:00 AM - 1:00 PM)

9:00 - 10:30 - File system, Permissions, Users & Groups [lecture]

10:30 - 10:45 - Break [break]

10:45 - 12:15 - Package Management, Process Management [demo]

12:15 - 1:00 - Knowledge check, use cases, and Q&A for Linux Basics [qa]

Afternoon Session: Linux Basics - Hands-on (1:00 PM - 5:00 PM)

1:00 - 2:30 - Linux Basics hands-on implementation, Linux Basics troubleshooting scenarios, Linux Basics best practices and review [lecture]

2:30 - 2:45 - Break [break]

2:45 - 4:00 - Lab: Navigate filesystem, manage users, and inspect service logs [lab]

4:00 - 5:00 - Jira: Update sprint board [jira]

Learning Objectives

- Understand Linux Basics concepts and terminology

- Use Ubuntu, Bash to complete guided exercises

- Apply Linux Basics in a real-world delivery scenario

- Connect the technical work to Agile/Jira delivery tracking

Jira Practice

- Update sprint board

- Create/update Epics, Stories, Tasks, Subtasks, acceptance criteria, and story points

- Move tasks across the sprint board and review progress with comments/time logs

Day 2: Networking Basics

Tools: curl + netstat + nmap | Jira Focus: Log time on stories

Morning Session: Networking Basics - Concepts (9:00 AM - 1:00 PM)

9:00 - 10:30 - OSI Model, TCP/IP, DNS [lecture]

10:30 - 10:45 - Break [break]

10:45 - 12:15 - HTTP/HTTPS, Ports & Protocols [demo]

12:15 - 1:00 - Knowledge check, use cases, and Q&A for Networking Basics [qa]

Afternoon Session: Networking Basics - Hands-on (1:00 PM - 5:00 PM)

1:00 - 2:30 - SSH, Networking Basics hands-on implementation, Networking Basics troubleshooting scenarios [lecture]

2:30 - 2:45 - Break [break]

2:45 - 4:00 - Lab: Diagnose network issues and configure SSH [lab]

4:00 - 5:00 - Jira: Log time on stories [jira]

Learning Objectives

- Understand Networking Basics concepts and terminology

- Use curl, netstat, nmap to complete guided exercises

- Apply Networking Basics in a real-world delivery scenario

- Connect the technical work to Agile/Jira delivery tracking

Jira Practice

- Log time on stories

- Create/update Epics, Stories, Tasks, Subtasks, acceptance criteria, and story points

- Move tasks across the sprint board and review progress with comments/time logs

Day 3: Docker Basics

Tools: Docker + Docker Hub | Jira Focus: Move cards to In Progress/Done

Morning Session: Docker Basics - Concepts (9:00 AM - 1:00 PM)

9:00 - 10:30 - Containers vs VMs, Images, Dockerfile [lecture]

10:30 - 10:45 - Break [break]

10:45 - 12:15 - Multi-stage Builds, Volumes [demo]

12:15 - 1:00 - Knowledge check, use cases, and Q&A for Docker Basics [qa]

Afternoon Session: Docker Basics - Hands-on (1:00 PM - 5:00 PM)

1:00 - 2:30 - Networking, Docker Basics hands-on implementation, Docker Basics troubleshooting scenarios [lecture]

2:30 - 2:45 - Break [break]

2:45 - 4:00 - Lab: Dockerize a Node.js/Python app [lab]

4:00 - 5:00 - Jira: Move cards to In Progress/Done [jira]

Learning Objectives

- Understand Docker Basics concepts and terminology

- Use Docker, Docker Hub to complete guided exercises

- Apply Docker Basics in a real-world delivery scenario

- Connect the technical work to Agile/Jira delivery tracking

Jira Practice

- Move cards to In Progress/Done

- Create/update Epics, Stories, Tasks, Subtasks, acceptance criteria, and story points

- Move tasks across the sprint board and review progress with comments/time logs

Day 4: Jenkins CI/CD

Tools: Jenkins + Maven + Nexus | Jira Focus: Add comments with build links

Morning Session: Jenkins CI/CD - Concepts (9:00 AM - 1:00 PM)

9:00 - 10:30 - Install & Setup, Freestyle Jobs, Declarative Pipeline [lecture]

10:30 - 10:45 - Break [break]

10:45 - 12:15 - Jenkinsfile, Agents [demo]

12:15 - 1:00 - Knowledge check, use cases, and Q&A for Jenkins CI/CD [qa]

Afternoon Session: Jenkins CI/CD - Hands-on (1:00 PM - 5:00 PM)

1:00 - 2:30 - Shared Libraries, Jenkins CI/CD hands-on implementation, Jenkins CI/CD troubleshooting scenarios [lecture]

2:30 - 2:45 - Break [break]

2:45 - 4:00 - Lab: Build, test, Docker push pipeline [lab]

4:00 - 5:00 - Jira: Add comments with build links [jira]

Learning Objectives

- Understand Jenkins CI/CD concepts and terminology
- Use Jenkins, Maven, Nexus to complete guided exercises
- Apply Jenkins CI/CD in a real-world delivery scenario
- Connect the technical work to Agile/Jira delivery tracking

Jira Practice

- Add comments with build links
- Create/update Epics, Stories, Tasks, Subtasks, acceptance criteria, and story points
- Move tasks across the sprint board and review progress with comments/time logs

Day 5: Terraform IaC

Tools: Terraform + AWS + S3 | Jira Focus: Sprint planning; review progress for Terraform IaC

Morning Session: Terraform IaC - Concepts (9:00 AM - 1:00 PM)

9:00 - 10:30 - Providers, Resources, Variables [lecture]

10:30 - 10:45 - Break [break]

10:45 - 12:15 - Outputs, State [demo]

12:15 - 1:00 - Knowledge check, use cases, and Q&A for Terraform IaC [qa]

Afternoon Session: Terraform IaC - Hands-on (1:00 PM - 5:00 PM)

1:00 - 2:30 - Modules, Workspaces, Remote Backend [lecture]

2:30 - 2:45 - Break [break]

2:45 - 4:00 - Lab: Provision VPC + EC2 + RDS with modules [lab]

4:00 - 5:00 - Jira: Sprint planning; review progress for Terraform IaC [jira]

Learning Objectives

- Understand Terraform IaC concepts and terminology
- Use Terraform, AWS, S3 to complete guided exercises
- Apply Terraform IaC in a real-world delivery scenario
- Connect the technical work to Agile/Jira delivery tracking

Jira Practice

- Sprint planning; review progress for Terraform IaC
- Create/update Epics, Stories, Tasks, Subtasks, acceptance criteria, and story points
- Move tasks across the sprint board and review progress with comments/time logs

Day 6: Kubernetes Fundamentals

Tools: kubectl + minikube + k9s | Jira Focus: Log time on stories

Morning Session: Kubernetes Fundamentals - Concepts (9:00 AM - 1:00 PM)

9:00 - 10:30 - Architecture, Pods, Deployments [lecture]

10:30 - 10:45 - Break [break]

10:45 - 12:15 - Services, Ingress [demo]

12:15 - 1:00 - Knowledge check, use cases, and Q&A for Kubernetes Fundamentals [qa]

Afternoon Session: Kubernetes Fundamentals - Hands-on (1:00 PM - 5:00 PM)

1:00 - 2:30 - ConfigMaps, Secrets, PV/PVC [lecture]

2:30 - 2:45 - Break [break]

2:45 - 4:00 - Lab: Deploy microservices app with Ingress [lab]

4:00 - 5:00 - Jira: Log time on stories [jira]

Learning Objectives

- Understand Kubernetes Fundamentals concepts and terminology

- Use kubectl, minikube, k9s to complete guided exercises

- Apply Kubernetes Fundamentals in a real-world delivery scenario

- Connect the technical work to Agile/Jira delivery tracking

Jira Practice

- Log time on stories

- Create/update Epics, Stories, Tasks, Subtasks, acceptance criteria, and story points

- Move tasks across the sprint board and review progress with comments/time logs

Day 7: AWS Core

Tools: AWS CLI + AWS Console + Terraform | Jira Focus: Move cards to In Progress/Done

Morning Session: AWS Core - Concepts (9:00 AM - 1:00 PM)

9:00 - 10:30 - IAM, EC2, S3 [lecture]

10:30 - 10:45 - Break [break]

10:45 - 12:15 - VPC, RDS [demo]

12:15 - 1:00 - Knowledge check, use cases, and Q&A for AWS Core [qa]

Afternoon Session: AWS Core - Hands-on (1:00 PM - 5:00 PM)

1:00 - 2:30 - Lambda, EKS, ECS [lecture]

2:30 - 2:45 - Break [break]

2:45 - 4:00 - Lab: Deploy 3-tier app on AWS with EKS + ALB [lab]

4:00 - 5:00 - Jira: Move cards to In Progress/Done [jira]

Learning Objectives

- Understand AWS Core concepts and terminology

- Use AWS CLI, AWS Console, Terraform to complete guided exercises

- Apply AWS Core in a real-world delivery scenario

- Connect the technical work to Agile/Jira delivery tracking

Jira Practice

- Move cards to In Progress/Done

- Create/update Epics, Stories, Tasks, Subtasks, acceptance criteria, and story points

- Move tasks across the sprint board and review progress with comments/time logs

Day 8: ELK Stack

Tools: ELK + Filebeat | Jira Focus: Add comments with build links

Morning Session: ELK Stack - Concepts (9:00 AM - 1:00 PM)

9:00 - 10:30 - Elasticsearch, Logstash, Kibana [lecture]

10:30 - 10:45 - Break [break]

10:45 - 12:15 - Filebeat, Log Pipelines [demo]

12:15 - 1:00 - Knowledge check, use cases, and Q&A for ELK Stack [qa]

Afternoon Session: ELK Stack - Hands-on (1:00 PM - 5:00 PM)

1:00 - 2:30 - Visualizations, ELK Stack hands-on implementation, ELK Stack troubleshooting scenarios [lecture]

2:30 - 2:45 - Break [break]

2:45 - 4:00 - Lab: Centralized log aggregation for containerized app [lab]

4:00 - 5:00 - Jira: Add comments with build links [jira]

Learning Objectives

- Understand ELK Stack concepts and terminology

- Use ELK, Filebeat to complete guided exercises

- Apply ELK Stack in a real-world delivery scenario

- Connect the technical work to Agile/Jira delivery tracking

Jira Practice

- Add comments with build links

- Create/update Epics, Stories, Tasks, Subtasks, acceptance criteria, and story points

- Move tasks across the sprint board and review progress with comments/time logs

Day 9: DevSecOps

Tools: Trivy + Snyk + AWS Secrets Manager | Jira Focus: Update sprint board

Morning Session: DevSecOps - Concepts (9:00 AM - 1:00 PM)

9:00 - 10:30 - SAST/DAST, Trivy, Snyk [lecture]

10:30 - 10:45 - Break [break]

10:45 - 12:15 - OPA/Gatekeeper, Secrets Manager [demo]

12:15 - 1:00 - Knowledge check, use cases, and Q&A for DevSecOps [qa]

Afternoon Session: DevSecOps - Hands-on (1:00 PM - 5:00 PM)

1:00 - 2:30 - IAM Policies, DevSecOps hands-on implementation, DevSecOps troubleshooting scenarios [lecture]

2:30 - 2:45 - Break [break]

2:45 - 4:00 - Lab: Add security scan stage to CI/CD pipeline [lab]

4:00 - 5:00 - Jira: Update sprint board [jira]

Learning Objectives

- Understand DevSecOps concepts and terminology

- Use Trivy, Snyk, AWS Secrets Manager to complete guided exercises

- Apply DevSecOps in a real-world delivery scenario

- Connect the technical work to Agile/Jira delivery tracking

Jira Practice

- Update sprint board

- Create/update Epics, Stories, Tasks, Subtasks, acceptance criteria, and story points

- Move tasks across the sprint board and review progress with comments/time logs

Day 10: Capstone Project + Certification Roadmap

Tools: All DevOps Tools | Jira Focus: Final sprint review, retrospective, release notes, and stakeholder demo

Morning Session: Capstone Project + Certification Roadmap - Concepts (9:00 AM - 1:00 PM)

9:00 - 10:30 - Requirements, Terraform Infra, Dockerize App [lecture]

10:30 - 10:45 - Break [break]

10:45 - 12:15 - CI/CD, K8s Deploy [demo]

12:15 - 1:00 - Knowledge check, use cases, and Q&A for Capstone Project + Certification Roadmap [qa]

Afternoon Session: Capstone Project + Certification Roadmap - Hands-on (1:00 PM - 5:00 PM)

1:00 - 2:30 - Monitoring, Security Scan, Demo [lecture]

2:30 - 2:45 - Break [break]

2:45 - 4:00 - Lab: Final project implementation, demo, retrospective, and certification roadmap review [lab]

4:00 - 5:00 - Jira: Final sprint review, retrospective, release notes, and stakeholder demo [jira]

Learning Objectives

- Understand Capstone Project + Certification Roadmap concepts and terminology

- Use All DevOps Tools to complete guided exercises

- Apply Capstone Project + Certification Roadmap in a real-world delivery scenario

- Connect the technical work to Agile/Jira delivery tracking

Jira Practice

- Final sprint review, retrospective, release notes, and stakeholder demo
- Create/update Epics, Stories, Tasks, Subtasks, acceptance criteria, and story points
- Move tasks across the sprint board and review progress with comments/time logs

Tools & Software

- Ubuntu
- Bash
- curl
- netstat
- nmap
- Docker
- Docker Hub
- Jenkins
- Maven
- Nexus
- Terraform
- AWS
- S3
- kubectl
- minikube
- k9s
- AWS CLI
- AWS Console
- ELK
- Filebeat
- Trivy
- Snyk
- AWS Secrets Manager
- All DevOps Tools

Hiring & Test Preparation

- Screening Test: 30-45 minute MCQ/practical test covering core DevOps concepts and tools
- Practical Assignment: hands-on task aligned with the client training requirement
- Mock Interview: trainer explains concepts, tools, troubleshooting approach, and delivery examples
- Trainer Demo: 15-20 minute sample teaching session with Q&A
- Evaluation Checklist: communication, technical depth, lab readiness, real-time examples, and client fit

Assessment Plan

- Daily knowledge check or lab review
- Weekly practical assignment for programs longer than 5 days
- Mid-program project review for 20+ day programs
- Final capstone demo and viva-style technical discussion

Primary Tools

- Ubuntu - used in hands-on labs and project delivery
- Bash - used in hands-on labs and project delivery
- curl - used in hands-on labs and project delivery
- netstat - used in hands-on labs and project delivery
- nmap - used in hands-on labs and project delivery
- Docker - used in hands-on labs and project delivery
- Docker Hub - used in hands-on labs and project delivery
- Jenkins - used in hands-on labs and project delivery
- Maven - used in hands-on labs and project delivery
- Nexus - used in hands-on labs and project delivery
- Terraform - used in hands-on labs and project delivery
- AWS - used in hands-on labs and project delivery

Project Management

- Jira - epics, stories, tasks, sprint board, reports
- Agile ceremonies - planning, review, retrospective

Certification Roadmap

- AWS Certified DevOps Engineer - Professional
- Certified Kubernetes Administrator (CKA)
- CKAD
- Terraform Associate
- GitHub Actions Certification
- Jenkins Certified Engineer
- Docker Certified Associate

Certification Guidance

Recommended certification path: AWS Certified DevOps Engineer - Professional, Certified Kubernetes Administrator (CKA), CKAD, Terraform Associate, GitHub Actions Certification, Jenkins Certified Engineer, Docker Certified Associate.

Trainer Notes

Generated by the Training TOC Agent from the curriculum knowledge base. Gemini may be used only for optional wording polish.